Population structure

Biologia della Conservazione 25/26

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Introduction

- Individuals rarely mate completely at random!
- Even within species, there's often geographically-restricted mating among individuals
- Individuals tend to mate with individuals from the same, or closely related sets of populations
- This form of non-random mating is called population structure and can have profound effects on the distribution of genetic variation within and among natural populations
- In most species individuals are scattered in local conglomerates that are called *demes* (*i.e.*, local populations or subpopulations)

- Summarizing population structure is often a key initial stage in population genetic and genomic analyses
- We can define inbreeding as having parents that are more closely related to each other than two individuals drawn at random from some reference population
- The question that naturally arises is: Which reference population should we use? While I might not look inbred in comparison to allele frequencies in Italy, where I am from, my parents certainly are not two individuals drawn at random from the world-wide population

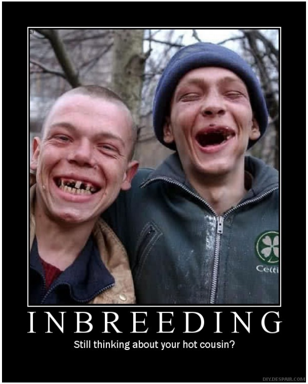


Figure 1: Inbreeding couple

Figure 2: Inbreeding couple

- There cannot be a class on inbreeding without these pictures!!

- In its most simple definition inbreeding is the mating between related individuals. An individual is considered inbred if its parents share a common ancestor
- In large population inbreeding can happen because of the non-random assortment of gametes
- In many tree species spatially closer trees are more likely to be related than trees further apart

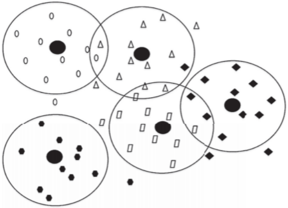


Figure 3: Image after Allendorf et al. (2022), pg. 178

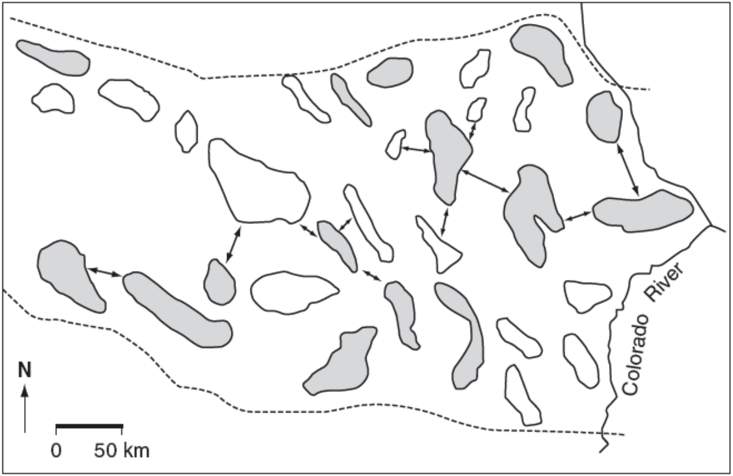


Figure 4: After Akçakaya, Mills, & Doncaster (2007)

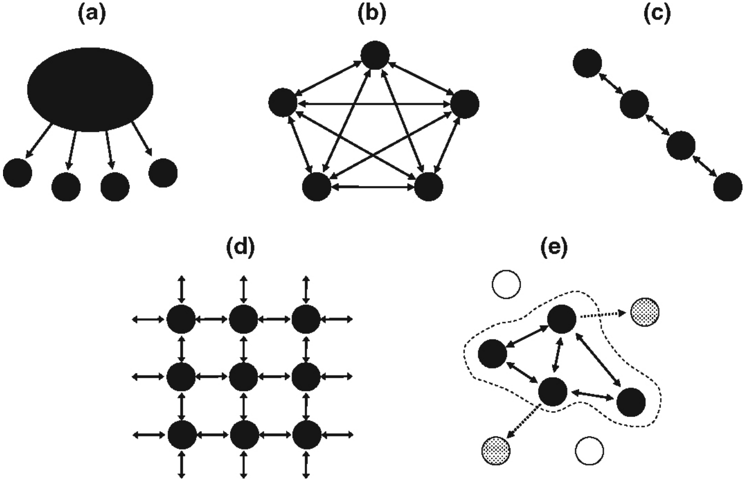


Figure 5: After Frankham, Ballou, & Briscoe (2010)

- A metapopulation is defined as a set of discrete populations (demes) of the same species, in the same general geographical area, potentially able to exchange individuals through migration, dispersal, or human-mediated movement (Akçakaya et al., 2007; Hanski & Simberloff, 1997)
- This subdivision (or structure) of populations can be natural and/or human driven
- Regardless of the cause, the existence of population structure in many species forces us to think at the genetic variation within a species at two primary levels:
    1. Genetic variation within local demes
    2. Genetic variation bewteen local demes

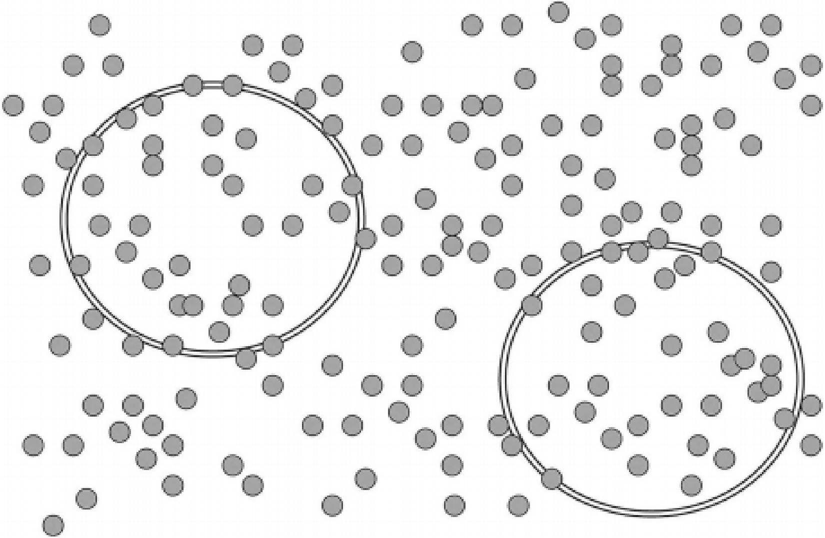


Figure 6: After Allendorf et al. (2022), p. 184

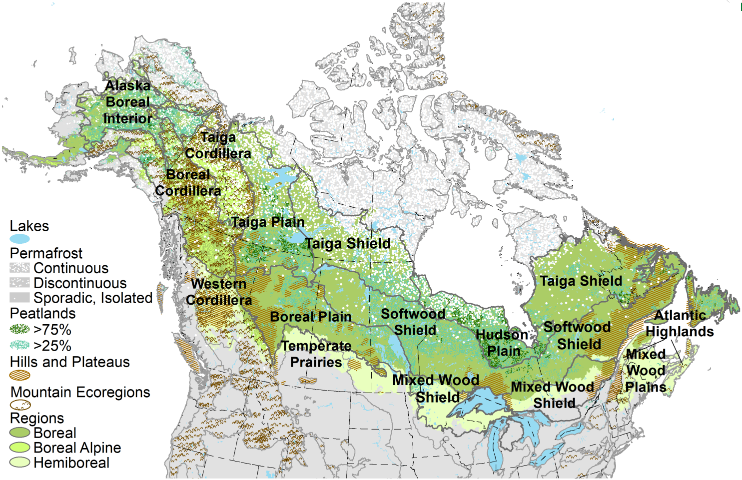


Figure 7: Boreal forest distibution in North America. Available here

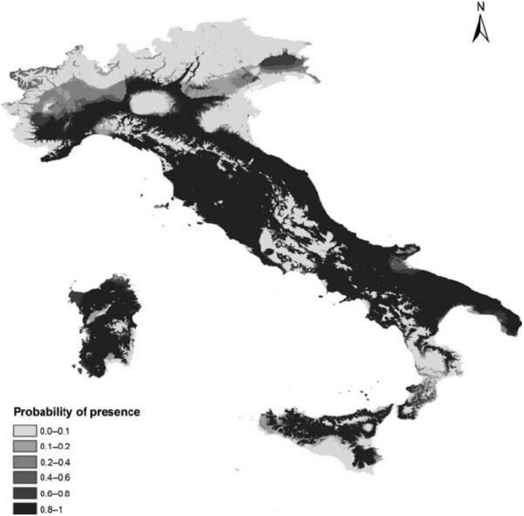


Figure 8: Maxent prediction of *Hystrix cristata* distribution After Mori, Sforzi, & Di Febbraro (2013)

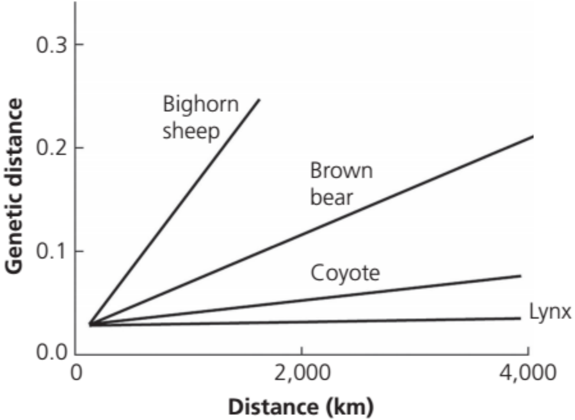


Figure 9: Individuals that are very distant from each other may present significant differences at the genetic level: isolation by distance. Modified from Forbes & Hogg (1999) with unpublished data from M.K. Schwartz (Allendorf et al., 2022).

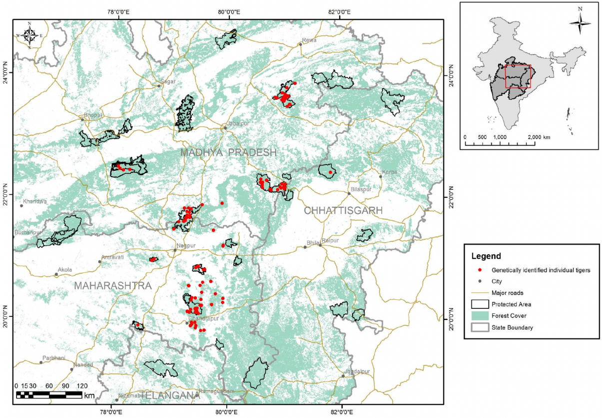


Figure 10: Describing the patterns and amount of differentiation among demes is of the uttermost importance to develop effective conservation strategies. Study area in Thatte, Joshi, Vaidyanathan, Landguth, & Ramakrishnan (2018). Genetically identified tigers as red dots

- Frankham et al. (2011) describe the risks of outbreeding depression when translocating individuals from genetically distinct populations to supplement imperiled ones

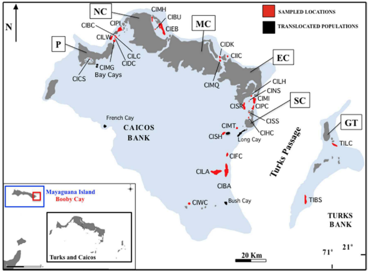


Figure 11: After Welch et al. (2017).

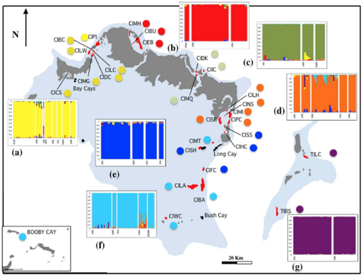


Figure 12: After Welch et al. (2017).

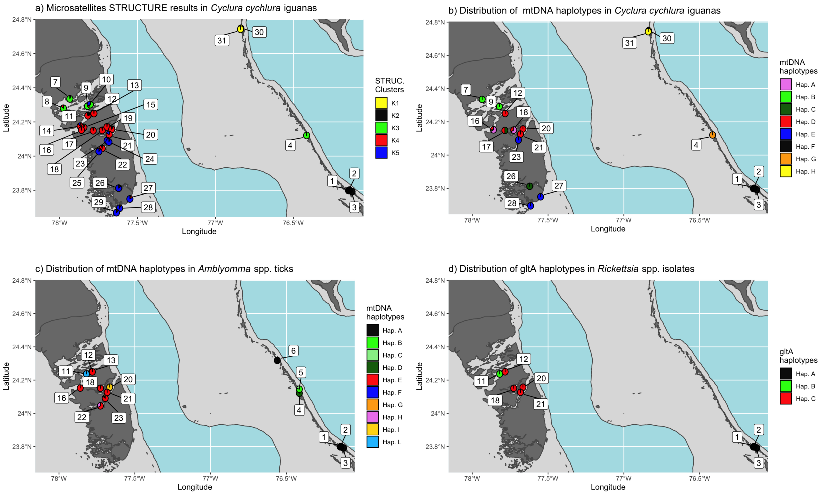


Figure 13: Patterns of differentiation in *Cyclura cychlura* populations. After Colosimo et al. (2021)

- Developing priorities for the conservation of a species often requires an understanding of adaptive genetic differentiation among populations (Allendorf et al., 2022)
- To understand how genetic variation is distributed at different loci within subdivided populations it is necessary to look at different interacting forces like gene flow, genetic drift and natural selection
- F statistics (Malécot, 1948; Wright, 1931, 1951) are undoubtedly the most used indexes to describe genetic differentiation in and between demes
- F statistics are, in short, inbreeding coefficients measuring different types of inbreeding at different levels

F statistics

- We recognize three primary coefficients: F_{IS}, F_{IT}, and F_{ST}
- F_{IS} measures Hardy–Weinberg departures within local demes
- F_{ST} measures divergence in allele frequencies among demes
- F_{IT} measures Hardy–Weinberg departures in the metapopulation due to F_{IS} (nonrandom mating within local demes) and F_{ST} (allele frequency divergence among demes)
- The original F statistics developed by wright focused on loci with only two alleles
- Nei (1977) extended these measures to multiple alleles and used a different notation to name what he called *analysis of gene diversity*: G_{IS}, G_{IT}, and G_{ST}. Often the two groups of indexes are used interchangeably

F_{IS}

- This is a measure of departure from HW proportions within local demes.

$$F_{IS} = 1 - \frac{H_O}{H_S} \tag{7.1}$$

- H_O is the observed heterozygosity averaged over all demes while H_S is the expected heterozygosity averaged over all demes. F_{IS} values can range from $-1 \leq 0 \leq 1$, with positive values indicating an excess of homozygotes and negative values indicating an excess of heterozygotes (Allendorf et al., 2022).

- Often called *fixation index*, it is a measure of genetic divergence among demes.

$$F_{ST} = 1 - \frac{H_S}{H_T} \tag{7.2}$$

- H_T as the expected HW heterozygosity if the metapopulation were panmictic, and H_S is the HW proportion of heterozygotes in each separate deme and then averaged over all demes. This measure varies from 0 (demes all have equal allele frequencies) to 1 (demes are all fixed for different alleles).

F_{IT}

- Measures the departure from HW proportions due to departures from HW proportions within local demes and divergence among demes (Allendorf et al., 2022).

$$F_{IT} = 1 - \frac{H_O}{H_T} \tag{7.3}$$

- H_O is the observed heterozygosity averaged over all demes while H_T is the expected HW heterozygosity if the metapopulation were panmictic.

- The fixation indexes described in the previous slides are related by the following expression:

$$F_{IT} = F_{IS} + F_{ST} - (F_{IS} * F_{ST}) \tag{7.4}$$

Population structure and habitat fragmentation

- Habitat fragmentation is a combination of two processes: reduction in total area, and the creation of separate islands
- Human induced fragmentation and habitat loss is recognized as the main cause of biodiversity loss (Baan, Alkemade, & Koellner, 2013)
- Habitat fragmentation causes an overall reduction in population size
- Habitat fragmentation also causes an overall reduction in gene flow among the patches it creates

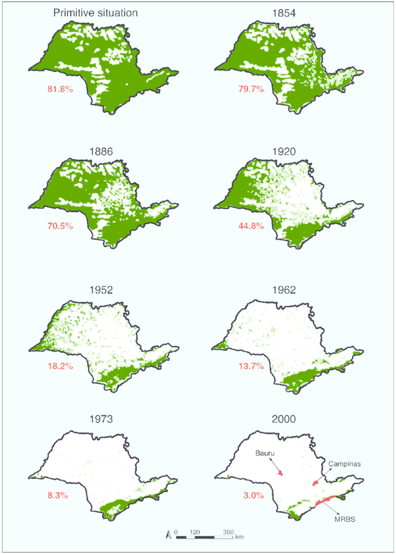


Figure 14: Fragmentation of Atlantic forest. After Nunes (2011)

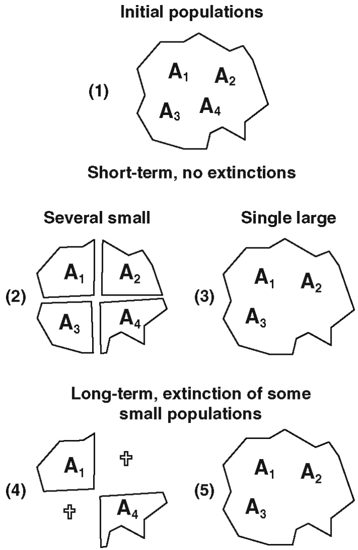


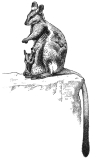
Figure 15: Single large (SL) VS several small (SS). After Frankham et al. (2010)

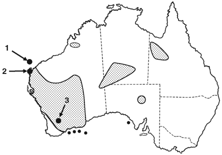
- Fragmentation happens in two steps:
  1. Initial genetic subdivision of the population
  2. Cumulative diversification through genetic drift, selection, inbreeding etc. etc. etc
- After an initial fragmentation, different fragments will have different initial allele frequencies. This diversification can be described by the binomial sampling variance ($\sigma_p^2 = \frac{pq}{2N}$) and the smaller the fragment, the higher the variance
- The degree of differentiation among fragments can be described by partitioning the overall inbreeding into components within and among populations (F statistics).
- We will look at some examples on estimating F statistics from genotype data.

Gene flow

- What is gene flow?
  - Movement of alleles between population due to migration.
- The effects of fragmentation on gene flow depend on:
  - Number of population fragments
  - Distribution of population sizes in the fragments
  - Distance between fragments
  - Dispersal ability of the species
  - Migration rates among fragments
  - Environment of the matrix among fragments
  - Times since fragmentation and extinction and recolonization rates across fragments

The black footed rock wallaby





Location	Prop. of polymorphic loci	Mean # of alleles	Average heterozygosity
Barrow Island (1)	0.1	1.2	0.05
Exmouth (2)	1.0	3.4	0.62
Wheatbelt (3)	1.0	4.4	0.56

- Data from Eldridge et al. (1999), images after Frankham et al. (2010).

Example - 1

- Populations with same allele frequencies. One with random mating the other one with inbreeding.

Population	A_1A_1	A_1A_2	A_2A_2	Allele Freq	F	He
1	0.25	0.5	0.25	$p = 0.5$ $q = 0.5$	0	0.5
2	0.4	0.2	0.4	$p = 0.5$ $q = 0.5$	0.6	0.5

- H_O observed heterozygosity averaged over all demes.
- H_S expected heterozygosity averaged over all demes.
- H_T expected heterozygosity of the metapopulation.
- $F_{IS} = 1 - \frac{H_O}{H_S}$; $F_{IT} = 1 - \frac{H_O}{H_T}$; $F_{ST} = 1 - \frac{H_S}{H_T}$.

Example - 2

- Random mating populations with different allele frequencies

Population	A_1A_1	A_1A_2	A_2A_2	Allele Freq	F	He
1	0.25	0.5	0.25	$p = 0.5$ $q = 0.5$	0	0.5
2	0.04	0.32	0.64	$p = 0.2$ $q = 0.8$	0	0.32

- H_O observed heterozygosity averaged over all demes
- H_S expected heterozygosity averaged over all demes
- H_T expected heterozygosity of the metapopulation
- $F_{IS} = 1 - \frac{H_O}{H_S}$; $F_{IT} = 1 - \frac{H_O}{H_T}$; $F_{ST} = 1 - \frac{H_S}{H_T}$

Example - 3

- Populations with different allele frequencies, one random mating and one with inbreeding

Population	A_1A_1	A_1A_2	A_2A_2	Allele Freq	F	He
1	0.25	0.5	0.25	$p = 0.5$ $q = 0.5$	0	0.5
2	0.14	0.13	0.74	$p = 0.2$ $q = 0.8$	0.6	0.32

- H_O observed heterozygosity averaged over all demes
- H_S expected heterozygosity averaged over all demes
- H_T expected heterozygosity of the metapopulation
- $F_{IS} = 1 - \frac{H_O}{H_S}$; $F_{IT} = 1 - \frac{H_O}{H_T}$; $F_{ST} = 1 - \frac{H_S}{H_T}$

Patterns of F_{ST}

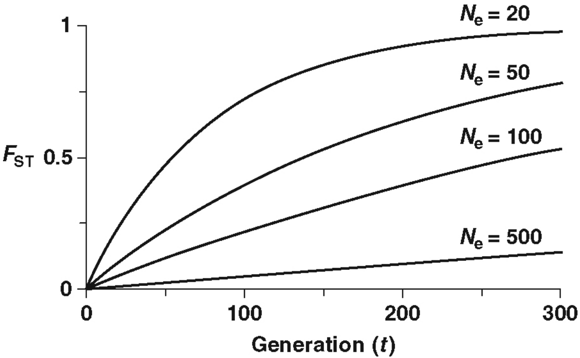
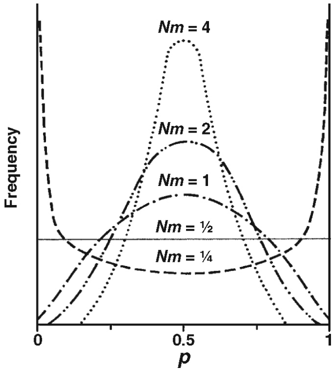


Figure 16: F_{ST} increases over generations in fragmented populations at a rate inversely dependent on population size. Values that are ≥ 0.15 indicate significant differentiation between fragments (Frankham et al., 2010).

Migration and gene flow

- Migration reduces the impact of fragmentation in a way that is dependent on the rate of gene flow
- When is gene flow sufficiently high to prevent the genetic impact of fragmentation?

- A single migrant per generation is considered sufficient to prevent differentiation between demes
- The migration rate m is the proportion of genes in a population derived from migrants per generation
- Nm is the number of migrants per generation (Frankham et al., 2010)



References I

Akçakaya, H. R., Mills, G., & Doncaster, C. P. (2007). The role of metapopulations in conservation.

Allendorf, F. W., Funk, W. C., Aitken, S. N., Byrne, M., Luikart, G., & Antunes, A. (2022). *Conservation and the Genomics of Populations*. Oxford University Press. Retrieved from https://doi.org/10.1093/oso/9780198856566.001.0001

Baan, L. de, Alkemade, R., & Koellner, T. (2013). Land use impacts on biodiversity in LCA: A global approach. *The International Journal of Life Cycle Assessment*, *18*(6), 1216–1230. Retrieved from https://link.springer.com/content/pdf/10.1007/s11367-012-0412-0.pdf

References II

Colosimo, G., Jackson, A. C., Benton, A., Varela-Stokes, A., Iverson, J., Knapp, C. R., & Welch, M. (2021). Correlated Population Genetic Structure in a Three-Tiered Host-Parasite System: The Potential for Coevolution and Adaptive Divergence. *Journal of Heredity*, *112*(7), 590–601. Retrieved from https://doi.org/10.1093/jhered/esab058

Eldridge, M. D. B., King, J. M., Loupis, A. K., Spencer, P. B. S., Taylor, A. C., Pope, L. C., & Hall, G. P. (1999). Unprecedented low levels of genetic variation and inbreeding depression in an island population of the black-footed rock-wallaby. *Conservation Biology*, *13*(3), 531–541. Retrieved from https://conbio.onlinelibrary.wiley.com/doi/abs/10.1046/j.1523-1739.1999.98115.x

References III

Forbes, S. H., & Hogg, J. T. (1999). Assessing population structure at high levels of differentiation: Microsatellite comparisons of bighorn sheep and large carnivores. *Animal Conservation*, *2*(3), 223–233. Retrieved from https://zslpublications.onlinelibrary.wiley.com/doi/abs/10.1111/j.1469-1795.1999.tb00068.x

Frankham, R., Ballou, J. D., & Briscoe, D. A. (2010). *Introduction to conservation genetics*. Cambridge University Press. Retrieved from https://doi.org/10.1017%2Fcbo9780511809002

Frankham, R., Ballou, J. D., Eldridge, M. D. B., Lacy, R. C., Ralls, K., Dudash, M. R., & Fenster, C. B. (2011). Predicting the probability of outbreeding depression. *Conservation Biology*, *25*(3), 465–475.

References IV

Hanski, I., & Simberloff, D. (1997). The metapopulation approach, its history, conceptual domain, and application to conservation. In I. Hanski & M. E. Gilpin (Eds.), *Metapopulation biology* (pp. 5–26). San Diego: Academic Press. Retrieved from https://www.sciencedirect.com/science/article/pii/B9780123234452500031

Malécot, G. (1948). *Les mathématiques de l'hérédité*. Masson. Retrieved from https://books.google.fi/books?id=g7FjAQAACAAJ

Mori, E., Sforzi, A., & Di Febbraro, M. (2013). From the apennines to the alps: Recent range expansion of the crested porcupineHystrix cristataL., 1758 (mammalia: Rodentia: Hystricidae) in italy. *Italian Journal of Zoology*, *80*(4), 469–480. Informa UK Limited.

References V

Nei, M. (1977). F-statistics and analysis of gene diversity in subdivided populations. *Annals of Human Genetics*, *41*(2), 225–233. Wiley. Retrieved from https://doi.org/10.1111%2Fj.1469-1809.1977.tb01918.x

Nunes, L. (2011). Landslides in são paulo, brazil: An integrated historical perspective. *Pages News*, *19*, 60–63.

Thatte, P., Joshi, A., Vaidyanathan, S., Landguth, E., & Ramakrishnan, U. (2018). Maintaining tiger connectivity and minimizing extinction into the next century: Insights from landscape genetics and spatially-explicit simulations. *Biological Conservation*, *218*, 181–191. Retrieved from https://www.sciencedirect.com/science/article/pii/S0006320717307346

References VI

Welch, M. E., Colosimo, G., Pasachnik, S. A., Malone, C. L., Hilton, J., Long, J., Getz, A. H., et al. (2017). Molecular variation and population structure in critically endangered Turks and Caicos Rock Iguanas: identifying intraspecific conservation units and revising subspecific taxonomy. *Conservation Genetics*, *18*(2), 479–493. Retrieved from http://link.springer.com/10.1007/s10592-016-0922-6

Wright, S. (1931). Evolution in Mendelian populations. *Bulletin of Mathematical Biology*, *52*(1-2), 241–295.

Wright, S. (1951). The genetical structure of populations. *Annals of eugenics*, *15*, 323–354.